

99 - Thread Minimum Cut

Used in G76 canned threading cycle, this setting sets a minimum amount of successive passes of the thread cut. Succeeding passes cannot be less than the value in this setting. The default value is .0010 inches.

101 - Feed Override -> Rapid

Pressing **[HANDLE FEED]**, with this setting **ON**, will cause the jog handle to affect both the feedrate and the rapid rate overrides. Setting 10 affects the maximum rapid rate. The rapid rate cannot exceed 100%. Also, **[+10% FEEDRATE]**, **[- 10% FEEDRATE]**, and **[100% FEEDRATE]** change the rapid and feed rate together.

102 - C Axis Diameter

This setting supports the C-Axis option.

This is a numeric entry. It is used to set the angular feed rate of the C-axis. Since the feed rate specified in a program is always inches per minute (or mm per minute), the control must know the diameter of the part being worked in the C-axis in order to compute the angular feed rate.

When this setting is set correctly, the surface feed rate on a spindle cut will be exactly the feed rate programmed into the control. See the C-Axis Section for more information.

103 - Cyc Start/Fh Same Key

The **[CYCLE START]** button must be pressed and held to run a program when this setting is **ON**. When **[CYCLE START]** is released, a feed hold is generated.

This setting cannot be turned on while Setting 104 is **ON**. When one of them is set to **ON**, the other automatically turns off.

104 - Jog Handle to SNGL BLK

The **[HANDLE JOG]** control can single-step through a program when this setting is **ON**. Reversing the **[HANDLE JOG]** control direction generates a feed hold.

This setting cannot be turned on while Setting 103 is **ON**. When one of them is set to **ON**, the other automatically turns off.

105 - TailStock Retract Distance

The distance from the rapid position the tailstock will retract when commanded. This setting should be a positive value.



NOTE:

This setting is in the User Positions tab under Settings

108 - Quick Rotary G28

If this setting is **ON**, the control returns the rotary axes to zero in +/-359.99 degrees or less.

For example, if the rotary unit is at +/-950.000 degrees and a zero return is commanded, the rotary table rotates +/-230.000 degrees to the home position if this setting is **ON**.



NOTE:

The rotary axis returns to the machine home position, not the active work coordinate position.

109 - Warm-Up Time in MIN.

This is the number of minutes (up to 300 minutes from power-up) during which the control applies the compensations specified in Settings 110-112.

Overview – When the machine is powered on, if Setting 109, and at least one of Settings 110, 111, or 112, are set to a nonzero value, the control gives this warning:

CAUTION! Warm up Compensation is specified!

Do you wish to activate

Warm up Compensation (Y/N)?

If you answer **Y** to the prompt, the control immediately applies the total compensation (Setting 110, 111, 112), and the compensation begins to decrease as the time elapses. For instance, after 50% of the time in Setting 109 has elapsed, the compensation distance is 50%.

To restart the time period, power the machine off and on, and then answer **YES** to the compensation query at start-up.



CAUTION:

Changing Setting 110, 111, or 112 while compensation is in progress can cause a sudden movement of up to 0.0044 inch.

110, 111, 112 - Warmup X, Y, Z Distance

Settings 110, 111, and 112 specify the amount of compensation (max = +/- 0.0020" or +/- 0.051 mm) applied to the axes. Setting 109 must have a value entered for settings 110-112 to have an effect.

113 - Tool Change Method

This setting is used for the TL-1 and TL-2 lathes. This setting selects how a tool change is executed.

A selection of **Auto** defaults to the automatic tool changer on the machine.

A selection of **Gang T1** allows you to implement a Gang T1 tool changer. A Gang T1 consists only of a change in tool offsets:

- T12 switches to tool 12 and uses the offset from tool 12
- T1213 switches to tool 12 and uses the offset from tool 13
- T1200 switches to tool 12 and uses no tool offset

A selection of **T1 Post** allows for a manual tool change operation. When a tool change is executed in a program, the machine will stop at a tool change and prompt you to load the tool. Load the spindle and press **[CYCLE START]** to continue the program.

114 - Conveyor Cycle (minutes)

Setting 114 Conveyor Cycle Time is the interval that the conveyor turns on automatically. For example, if setting 114 is set to 30, the chip conveyor turns on every half an hour.

On-time should be set no greater than 80% of cycle time. Refer to Setting 115 on page **388**.

NOTE: *The **[CHIP FWD]** button (or M31) starts the conveyor in the forward direction and starts the cycle.*

*The **[CHIP STOP]** button (or M33) stops the conveyor and cancels the cycle.*

115 - Conveyor On-time (minutes)

Setting 115 Conveyor On-Time is the amount of time the conveyor runs. For example, if setting 115 is set to 2, the chip conveyor runs for 2 minutes, then turns off.

On-time should be set no greater than 80% of cycle time. Refer to Setting 114 Cycle Time on page **388**.

NOTE: *The **[CHIP FWD]** button (or M31) starts the conveyor in the forward direction and starts the cycle.*

*The **[CHIP STOP]** button (or M33) stops the conveyor and cancels the cycle.*

117 - G143 Global Offset (VR Models Only)

This setting is provided for customers who have several 5-axis Haas mills and want to transfer the programs and tools from one to another. The pivot-length difference is entered into this setting, and it is applied to the G143 tool length compensation.

118 - M99 Bumps M30 CNTRS

When this setting is **ON**, an **M99** adds one to the **M30** counters (these are visible after pressing **[CURRENT COMMANDS]**).



NOTE:

M99 only increases the counters as it occurs in a main program, not in a sub-program.

119 - Offset Lock

Turning the setting **ON** does not allow the values in the Offset display to be altered. However, programs that alter offsets with macros or **G10** are permitted to do so.

120 - Macro Var Lock

Turning this setting **ON** does not allow the macro variables to be altered. However, programs that alter macro variables can do so.

130 - Tap Retract Speed

This setting affects the retract speed during a tapping cycle (The mill must have the Rigid Tapping option). Entering a value, such as 2, commands the mill to retract the tap twice as fast as it went in. If the value is 3, it retracts three times as fast. A value of 0 or 1 has no effect on the retract speed.

Entering a value of 2 is the equivalent of using a **T** address code value of 2 for **G84** (tapping canned cycle). However, specifying a **T** code for a rigid tap overrides Setting 130.

131 - Auto Door

This setting supports the Auto Door option. It should be set to **ON** for machines with an autodoor. Also see **M85/M86** (Autodoor Open/Close M-codes).



NOTE:

The M-codes work only while the machine receives a cell-safe signal from a robot. For more information, contact a robot integrator.

The door closes when **[CYCLE START]** is pressed and opens when the program reaches an **M00**, **M01** (with Optional Stop turned on), **M02** or **M30** and the spindle has stopped turning.

133 - Repeat Rigid Tap

This setting (Repeat Rigid Tap) ensures that the spindle is oriented during tapping so that the threads line up when a second tapping pass is programmed in the same hole.



NOTE:

*This setting must be **ON** when a program commands peck tapping.*

142 - Offset Chng Tolerance

This setting is intended to prevent operator errors. It generates a warning message if an offset is changed by more than the setting's value, 0 to 3.9370 inches (0 to 100 mm). If you change an offset by more than the entered amount (either positive or negative), the control prompts: *XX changes the offset by more than Setting 142! Accept (Y/N)?*

Press **[Y]** to continue and update the offset. Press **[N]** to reject the change.

143 - Machine Data Collection Port

When this setting has a non-zero value, it defines the network port that the control uses to send machine data collection information. If this setting has a value of zero, the control does not send machine data collection information.

144 - Feed Override->Spindle

This setting is intended to keep the chip load constant when an override is applied. When this setting is **ON**, any feedrate override is also applied to the spindle speed, and the spindle overrides are disabled.

145 - Tailstock At Part for Cycle Start

When Setting 145, Tailstock at Part for **[CYCLE START]** is **OFF**, the machine behaves as before. When this setting is **ON**, the tailstock must be pressing against the part at the moment **[CYCLE START]** is pressed or alarm 9109 TAILSTOCK NOT IN PART HOLD POSITION is displayed and the program will not start.

155 - Load Pocket Tables

This setting is used when a software upgrade is performed and/or memory has been cleared and/or the control is re-initialized. In order to replace the contents of the side-mount tool changer pocket tool table with the data from the file, the setting must be **ON**.

If this setting is **OFF** when loading an Offset file from a hardware device, the contents of the **Pocket Tool** table is unaltered. Setting 155 automatically defaults to **OFF** when the machine is turned on.

156 - Save Offsets with Program

When this setting is **ON**, the control includes the offsets in the program file when you save it. The offsets appear in the file before the final % sign, under the heading `O999999`.

When you load the program back into memory, the control prompts *Load Offsets (Y/N?)*. Press **Y** if you want to load the saved offsets. Press **N** if you do not want to load them.

158, 159, 160 - X, Y, Z Screw Thermal COMP%

These settings can be set from -30 to +30 and adjust the existing screw thermal compensation by -30% to +30% accordingly.

162 - Default To Float

When this setting is **ON**, the control will interpret the interger code as if it had a decimal point. When the setting is **OFF**, values after address codes that do not include decimal points are taken as machinist's notation; for example, thousandths or ten-thousandths.

	Value entered	With Setting Off	With Setting On
In Inch mode	X-2	X-.0002	X-2.
In MM mode	X-2	X-.002	X-2.

This feature applies to these address codes:

X, Y, Z, A, B, C, E, I, J, K, U, W

Including A and D except when:

- the **A** value (tool angle) is in a **G76** block. If a **G76 A** value containing a decimal point is found during program execution, Alarm 605 - Invalid Tool Nose Angle is generated.
- the **D** value is in a **G73** block.



NOTE:

This setting affects the interpretation of all programs. It does not alter the effect of Setting 77 - Scale Integer F.

163 - Disable .1 Jog Rate

This setting disables the highest jog rate. If the highest jog rate is selected, the next lower rate is automatically selected instead.

165 - Main Spindle SSV Variation (RPM)

Specifies the amount by which to allow the RPM to vary above and below the commanded value during use of the Spindle Speed Variation feature. This must be a positive value.

166 - Main Spindle SSV Cycle

Specifies the duty cycle, or the rate of change of the Main Spindle Speed. This must be a positive value.

191 - Default Smoothness

This setting's value of **ROUGH**, **MEDIUM**, or **FINISH** sets the default smoothness and a maximum corner rounding factor. The control uses this default value unless a **G187** command overrides the default.

196 - Conveyor Shutoff

This specifies the amount of time to wait without activity prior to turning off the chip conveyor. Units are minutes.

197 - Coolant Shutoff

This setting is the amount of time to wait without activity before Coolant flow stops. Units are minutes.

199 - Backlight Timer

This setting is the time in minutes after which the machine display backlight turns off when there is no input at the control (except in JOG, GRAPHICS, or SLEEP mode or when an alarm is present). Press any key to restore the screen (**[CANCEL]** is preferred).

216 - Servo and Hydraulic Shutoff

This setting specifies the duration of idle time, in seconds, before Power Save Mode starts. Power Save Mode shuts down all servo motors and hydraulic pumps. The motors and pumps start up again when needed (axis/spindle motion, program execution, etc.).

232 - G76 Default P Code

The default **P** code value to use when a **P** code does not exist in a **G76** line, or when the **P** code used has a value less than 1 or greater than 4. Possible values are **P1**, **P2**, **P3**, or **P4**.

238 - High Intensity Light Timer (minutes)

Specifies the duration in minutes that the High Intensity Light option (HIL) remains turned on when activated. The light turns on when the door is opened and the work light switch is on. If this value is zero, then the light will remain turned on while the doors are open.

239 - Worklight Off Timer (minutes)

Specifies the amount of time in minutes after which the work light will turn off automatically if there are no key presses or **[HANDLE JOG]** changes. If a program is running when the light turns off, the program will continue running.

240 - Tool Life Warning

This value is a percentage of tool life. When tool wear reaches this threshold percentage, the control displays a Tool Wear Warning icon.

241 - Tailstock Hold Force

Force to apply to a part by the servo tailstock (ST-40/45, ST-40L/40L and ST-50/55 only). Unit is pounds-force in standard mode and Newton in metric mode, as per Setting 9.

T9.1: Servo Tailstock Specifications

Min Thrust (Programmable Minimum)	Max Thrust (Programmable Maximum)
1000 lb / 4448 N	4500 lb / 20017 N

242 - Air Water Purge Interval (minutes)

This setting specifies the interval, in minutes, between condensate purges from the system air reservoir.

243 - Air Water Purge On-Time (seconds)

This setting specifies the duration, in seconds, of condensate purges from the system air reservoir.

245 - Hazardous Vibration Sensitivity

This setting has (3) levels of sensitivity for the hazardous vibration accelerometer in the machine's control cabinet: **Normal**, **Low**, or **Off**. The value defaults to **Normal** at each machine power-up.

You can see the current g force reading on the **Gauges** page in **Diagnostics**.

Depending on the machine, vibration is considered hazardous when it exceeds 600 - 1,400 g. At or above the limit, the machine gives an alarm.

If your application tends to cause vibration, you can change Setting 245 to a lower sensitivity to prevent nuisance alarms.

247 - Simultaneous XYZ Motion in Tool Change

Setting 247 defines how the axes move during a tool change. If Setting 247 is **OFF**, the Z Axis retracts first, followed by X- and Y-Axis motion. This feature can be useful in avoiding tool collisions for some fixture configurations. If Setting 247 is **ON**, the axes move simultaneously. This may cause collisions between the tool and the workpiece, due to B- and C-Axis rotations. It is strongly recommended that this setting remain **OFF** on the UMC-750, due to the high potential for collisions.

250 - Mirror Image C Axis

This is an **ON/OFF** setting. When it is **OFF**, axis motions occur normally. When it is **ON**, C-Axis motion may be mirrored (or reversed) around the work zero point. Also, see G101 and Settings 45, 46, 47, 48, and 80.

251 - Subprogram Search Location

This setting specifies the directory to search for external subprograms when the subprogram is not in the same directory as the main program. Also, if the control cannot find an **M98** subprogram, the control looks here. Setting 251 has (3) options:

- **Memory**
- **USB Device**
- **Setting 252**

For the **Memory** and **USB Device** options, the subprogram must be in the root directory of the device. For the **Setting 252** selection, Setting 252 must specify a search location to use.



NOTE:

*When you use **M98**:*

- The **P** code (**nnnnn**) is the same as the program number (**Onnnnn**) of the subprogram.
- If the subprogram is not in memory, the file name must be **Onnnnn.nc**. The file name must contain the **O**, leading zeros and **.nc** for the machine to find the subprogram.

252 - Custom Subprogram Search Location

This setting specifies the subprogram search locations when Setting 251 is set to **Setting 252**. To make changes to this setting, highlight Setting 252 and press the **[RIGHT]** cursor. The Setting 252 popup explains how to delete and add search paths and lists existing search paths.

To delete a search path:

1. Highlight the path listed in the Setting 252 popup.
2. Press **[DELETE]**.

If there is more than one path to delete, repeat steps 1 and 2.

To set a new path:

1. Press **[LIST PROGRAM]**.
2. Highlight the directory to add.
3. Press **[F3]**.
4. Select **Setting 252 add** and press **[ENTER]**.

To add another path, repeat steps 1 through 4.



NOTE:

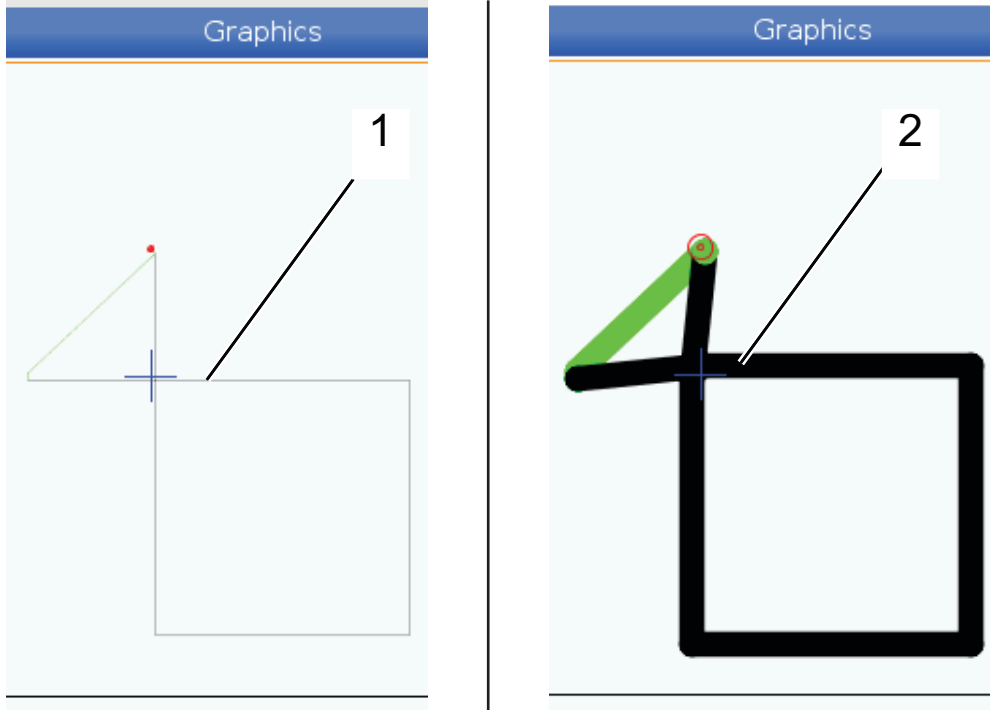
When you use M98:

- The **P** code (nnnnn) is the same as the program number (Onnnnn) of the subprogram.
- If the subprogram is not in memory, the file name must be Onnnnn .nc. The file name must contain the O, leading zeros and .nc for the machine to find the subprogram.

253 - Default Graphics Tool Width

If this setting is **ON**, Graphics mode uses default tool width (a line) [1]. If this setting is **OFF**, Graphics mode uses the Tool Offset Diameter Geometry specified in the **Tool Offsets** table as the graphics tool width [2].

F9.10: Graphics Display with Setting 253 On [1] and Off [2].



261 - DPRNT Store Location

DPRNT is a macro function that lets the machine control communicate with external devices. The Next-Generation Control (NGC) lets you output DPRNT statements over a TCP network, or to a file.

Setting 261 lets you specify where the DPRNT statement output goes:

- **Disabled** - The control does not process DPRNT statements.
- **File** - The control outputs DPRNT statements to the file location specified in setting 262.
- **TCP Port** - The control outputs DPRNT statements to the TCP port number specified in setting 263.

262 - DPRNT Destination File Path

DPRNT is a macro function that lets the machine control communicate with external devices. The Next-Generation Control (NGC) lets you output DPRNT statements to a file, or over a TCP network.

If setting 261 is set to **File**, setting 262 lets you specify the file location where the control sends DPRNT statements.

263 - DPRNT Port

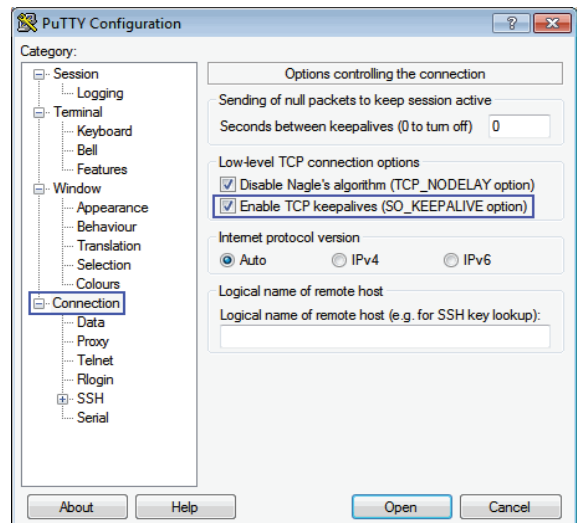
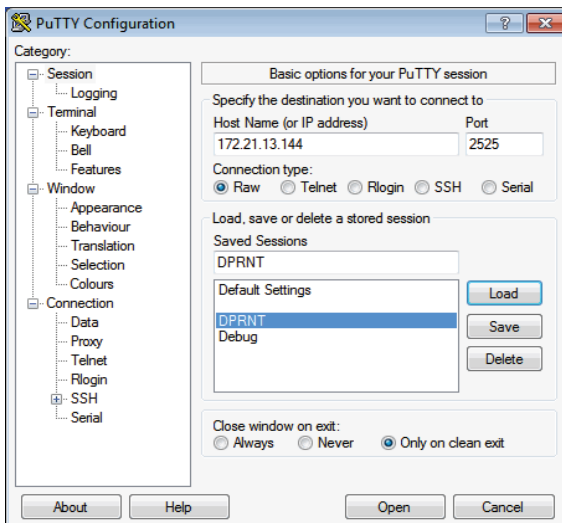
DPRNT is a macro function that lets the machine control communicate with external devices. The Next-Generation Control (NGC) lets you output DPRNT statements over a TCP network.

If setting 261 is set to **TCP Port**, setting 263 lets you specify the TCP port where the control sends DPRNT statements. On the PC, you can use any terminal program that supports TCP.

Use the port value along with the machine's IP address in the terminal program to connect to the machine's DPRNT stream. For example, if you use the terminal program PUTTY:

1. In the basic options section, type the machine's IP address and the port number in Setting 263.
2. Select the Raw or Telnet connection type.
3. Click "Open" to start the connection.

F9.11: PUTTY can save these options for subsequent connections. To keep the connection open, select "Enable TCP keepalives" in the "Connection" options.



To check the connection, type ping in the PUTTY terminal window and press enter. The machine sends a ping message if the connection is active. You can establish up to (5) simultaneous connections at a time.

264 - Autofeed Step Up

While autofeed is active, this setting defines the percentage amount by which the feedrate increments after tool overload stops.

265 - Autofeed Step Down

When autofeed is active, this setting defines the percentage amount by which the feedrate decrements during a tool overload.

266 - Autofeed Minimum Override

This setting defines the minimum percentage to which autofeed can reduce the feedrate.

267 - Exit Jog Mode after Idle Time

This setting defines the maximum duration, in minutes, that the control remains in jog mode with no axis motion or keyboard activity. After this duration, the control automatically changes to **MDI** mode. A value of zero disables this automatic change to **MDI** mode from jog mode.

268 - Second Home Position X

This setting defines the X-Axis position for second home, in inches or millimeters. The value is limited by the travel limits for the specific axis.

Press the **[ORIGIN]** button to set this setting to inactive or set the complete group to inactive.



NOTE:

*This setting is in the **User Positions** tab under **Settings**. Refer to the tab description on page **424** for more information.*



CAUTION:

Incorrectly set user positions can cause machine crashes. Set user positions with caution, especially after you have changed your application in some way (new program, different tools, etc.). Verify and change each axis position separately.

269 - Second Home Position Y

This setting defines the Y-Axis position for second home, in inches or millimeters. The value is limited by the travel limits for the specific axis.

Press the **[ORIGIN]** button to set this setting to inactive or set the complete group to inactive.



NOTE:

*This setting is in the **User Positions** tab under **Settings**. Refer to the tab description on page **424** for more information.*



CAUTION:

Incorrectly set user positions can cause machine crashes. Set user positions with caution, especially after you have changed your application in some way (new program, different tools, etc.). Verify and change each axis position separately.

270 - Second Home Position Z

This setting defines the Z-Axis position for second home, in inches or millimeters. The value is limited by the travel limits for the specific axis.

Press the **[ORIGIN]** button to set this setting to inactive or set the complete group to inactive.



NOTE:

*This setting is in the **User Positions** tab under **Settings**. Refer to the tab description on page **424** for more information.*



CAUTION:

Incorrectly set user positions can cause machine crashes. Set user positions with caution, especially after you have changed your application in some way (new program, different tools, etc.). Verify and change each axis position separately.

276 - Workholding Input Number

This setting specifies the input number to monitor for workholding fixture clamping. If the control receives a spindle start command while this input indicates that the workholding is not clamped, the machine gives an alarm.

277 - Axis Lubrication Interval

This setting defines the interval, in hours, between cycles for the axis lubrication system. The minimum value is 1 hour. The maximum value is between 12 and 24 hours, depending on the machine model.

281 - Chuck Foot Pedal Lockout

This is an **ON/OFF** setting. When it is **OFF**, the chuck foot pedal operates normally. When it is **ON**, any action at the foot pedal is ignored by the control.

282 - Main Spindle Chuck Clamping

This setting determines the Main Spindle chuck clamping direction. Set to O.D., the chuck is considered clamped when the jaws are moved to the spindle center. Set to I.D., the chuck is considered clamped when the jaws are moved away from the spindle center.

283 - Main Spindle Chuck Unclamp RPM

This setting determines the maximum main spindle speed for unclamping the chuck. The RPM at which the chuck will not operate. If the main spindle is spinning faster than this value the chuck will not open. If the main spindle is spinning slower than this value the chuck will open.

284 - Cycle Start Allowed With Chuck Unclamped

This setting allows Cycle Start to function with the chuck unclamped.

285 - X Diameter Programming

This setting sets diameter for programming. When this setting is set to TRUE, it interprets inputs as diameter instead of radius.

286 - Canned Cycle Cut Depth

Used with canned cycles *G71* and *G72*, this setting specifies the incremental depth for each pass during rough cutting. It is used if the programmer does not specify a *D* code. The default value is 0.100 inches.

287 - Canned Cycle Retraction

Used with canned cycles *G71* and *G72*, this setting specifies the retraction amount after a roughing cut. It represents the tool to material clearance as the tool returns for another pass.

289 - Thread Finish Allowance

Used in *G76* canned threading cycle, this setting specifies how much material is left on the thread for the final pass of the cycle.

291 - Main Spindle Speed Limit

This setting defines a top speed for the main spindle. When this setting has a nonzero value, the spindle will never exceed the designated speed.

292 - Door Open Spindle Speed Limit

This setting specifies the maximum spindle speed allowed while the machine door is open

306 - Minimum Chip Clear Time

This setting specifies the minimum amount of time, in seconds, that the spindle remains at “chip clean speed” (the spindle RPM designated in a canned cycle E command). Add time to this setting if your commanded chip clean cycles do not completely remove the chips from the tool.

313, 314, 315 - Max User Travel Limit X, Y, Z

This setting lets you define a custom travel limit position for the X, Y, and Z axis.

Press the **[ORIGIN]** button to set this setting to inactive or set the complete group to inactive.



NOTE:

*This setting is in the **User Positions** tab under **Settings**. Refer to the tab description on page **424** for more information.*

319 - VDI Spindle Center Line X

This setting lets you define the machine position which aligns the center of the VDI tool holder with the center of the spindle.



NOTE:

*This setting is in the **User Positions** tab under **Settings***

320 - BOT Spindle Center Line X

This setting lets you define the machine position which aligns the center of the BOT tool holder with the center of the spindle.



NOTE:

*This setting is in the **User Positions** tab under **Settings***

321 - Spindle Center Line Y

This setting lets you define the machine position which aligns the center of the tool holders with the center of the spindle for the Y Axis.



NOTE:

*This setting is in the **User Positions** tab under **Settings***

322 - Foot Pedal Tailstock Alarm

When **M21** is used to move the tailstock to the hold point and hold a part, the control generates an alarm if a part is not found and the hold point is reached. Setting 322 can be switched to **ON** and an alarm is generated when the foot pedal is used to move the tailstock to the hold point and no part is found.

323 - Disable Notch Filter

When this setting is **ON**, the notch filter values are set to zero. When this setting is **OFF**, it uses the default values of the machine as set defined by parameters. Turning this setting **ON** will improve circular accuracy and turning **OFF** will improve surface finish.



NOTE:

You must cycle power for this setting to take effect.

325 - Manual Mode Enabled

Turning this setting **ON** allows the axes to be jogged without zero returning the machine (finding machine home).

The jog limits imposed by setting 53 Jog W/O Zero Return will not apply. The jog rate will be defined by the eWheel switch or the jog rate buttons (if the eWheel is not connected).

With this setting **ON** you can perform tool changes using the **[ATC FWD]** or **[ATC REV]** buttons.

When turning this setting **OFF** the machine will operate as normal, and will require to be zero return.

326 - Graphics X Zero Location

This setting locates the top of the zoom window relative to the machine X zero position (see the Graphics section). Its default is zero.

327 - Graphics Z Zero Location

This setting locates the top of the zoom window relative to the machine Z zero position (see the Graphics section). Its default is zero.

328 - eHandwheel Rapid Limit

This setting allows you to limit how fast the eHandwheel moves when you press and hold down the rapid button. A value of zero disables the button.

329 - Main Spindle Jog Speed

This setting determines the spindle RPM for the spindle Jog key.

330 - MultiBoot Selection Time out

This is a simulator only setting. When a simulator is powered on, it displays a screen from where different simulator models can be chosen. This setting sets how long that screen is shown. If the user does nothing before the time expires, the software will load the last active simulator configuration.

331 - Sub Spindle Jog Speed

This setting determines the spindle rpm for the spindle jog key.

332 - Tailstock Foot Pedal LockOut

This is an **ON/OFF** setting. When it is **OFF**, the tailstock foot pedal operates normally. When it is **ON**, any action at the tailstock foot pedal is ignored by the control.

333, 334 - Probe Offset Z+, Z-

These settings are used to define the displacement and size of the ATP. These four settings (59, 60, 333, 334) specify the travel distance and direction from where the probe is triggered to where the actual sensed surface is located.

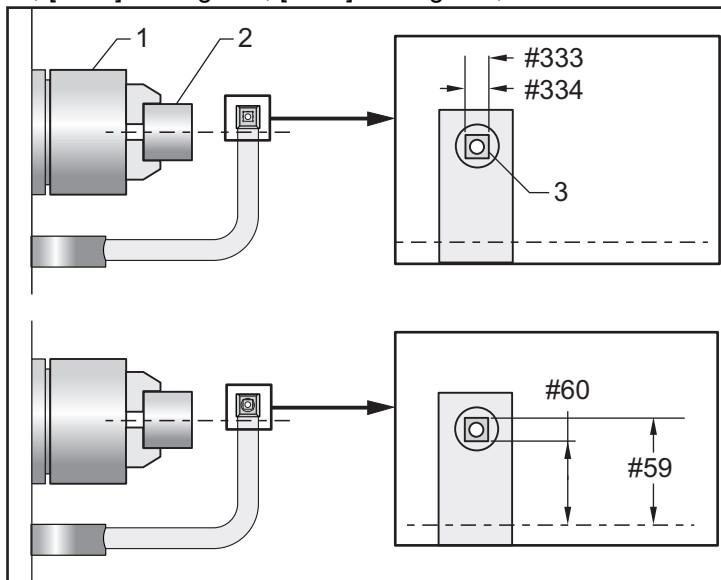
For more information on how to calibrate the ATP refer to page **179**.

These settings are used by the G31 code. The values entered for each setting must be positive numbers.

Macros can be used to access these settings, see the Macro section for more information.

F9.12:

59/60/333/334 Tool Probe Offset:[1] Chuck, [2] Part, [3] Probe, [#59] Setting 59, [#60] Setting 60, [#333] Setting 333, [#334] Setting 334,



335 - Linear Rapid Mode

This setting can be set to one of two modes. The description of these modes is as follows:

NONE The individual axis rapid to their endpoints independently of each other.

LINEAR (XYZ) The XYZ axes, when commanded to rapid, move linearly through 3D space. All other axis rapid with independent speeds/accelerations.



NOTE:

All modes cause a program to run in the same amount of time (no increase or decrease in execution time).

336 - Bar Feeder Enable

This setting turns on the Bar Feeder tab in **[CURRENT COMMANDS]** under the **Devices** Tab. Use this page to set up the Bar Feeder.

337, 338, 339 - Safe Tool Change Location X, Y, Z

These settings let you define a safe position for the X, Y and Z axis at a tool change command, before the axes go to their final tool change positions. Use this position to avoid collisions with fixtures, tailstock, and other potential obstacles. The control uses this position for every tool change, no matter how it is commanded (M06, **[NEXT TOOL]**, etc.)



CAUTION:

Incorrectly set user positions can cause machine crashes. Set user positions with caution, especially after you have changed your application in some way (new program, different tools, etc.). Verify and change each axis position separately.

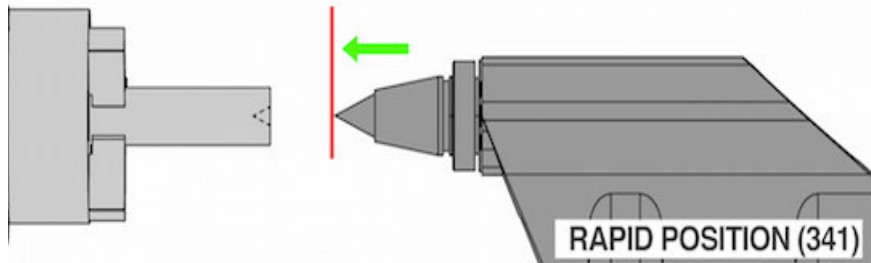
340 - Chuck Clamp Delay Time

The dwell time that is allowed after clamping the chuck (an M10 command). Program execution will not continue until this time has expired.

341 - Tailstock Rapid Position

This is the point where the tailstock will change from rapid to feed movement when moving toward the part. This setting should be a negative value.

F9.13: Tailstock Rapid Position



NOTE:

This setting is in the User Positions tab under Settings

342 - Tailstock Advance Distance

This setting is the distance from the Tailstock Rapid Position to a point just inside the part.

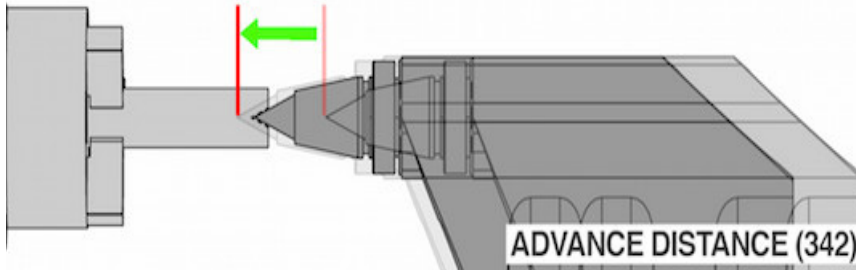
To determine the value for this setting:

- Jog the tailstock to the part face

- Subtract current position from the Retract Position to get the distance from the Retract Position to the part face
- Then add 0.375 – 0.500" (9.5 – 12.7mm)

The machine will use this setting to calculate a target position inside the part, relative to the Rapid Position (Setting 341).

F9.14: Tailstock Advance Distance



NOTE:

This setting is in the User Positions tab under Settings

343 - Sub Spindle SSV Variation (RPM)

Specifies the amount by which to allow the RPM to vary above and below the commanded value during use of the Sub Spindle Speed Variation feature. This must be a positive value.

344 - Sub Spindle SSV Cycle

Specifies the duty cycle, or the rate of change of the Sub Spindle Speed. This must be a positive value.

345 - Sub Spindle Chuck Clamping

This setting determines the Sub Spindle chuck clamping direction. Set to O.D., the chuck is considered clamped when the jaws are moved to the sub spindle center. Set to I.D., the chuck is considered clamped when the jaws are moved away from the sub spindle center.

346 - Sub Spindle Chuck Unclamp RPM

This setting determines the maximum sub spindle speed for unclamping the chuck. The RPM at which the chuck will not operate. If the sub spindle is spinning faster than this value the chuck will not open. If the sub spindle is spinning slower than this value the chuck will open.

347 - Live Tooling SSV Variation (RPM)

Specifies the amount by which to allow the RPM to vary above and below the commanded value during use of the Live Tooling Speed Variation feature. This must be a positive value.

348 - Live Tooling SSV Cycle

Specifies the duty cycle, or the rate of change of the Live Tooling Speed. This must be a positive value.

349 - Live Tooling Chuck Clamping

This setting determines the Live Tooling clamping direction. Set to O.D., the chuck is considered clamped when the jaws are moved to the Live Tooling center. Set to I.D., the chuck is considered clamped when the jaws are moved away from the Live Tooling center.

350 - Live Tooling Chuck Unclamp RPM

This setting determines the maximum Live Tooling speed for unclamping the chuck. The RPM at which the chuck will not operate. If the Live Tooling is spinning faster than this value the chuck will not open. If the Live Tooling is spinning slower than this value the chuck will open.

352 - Live Tooling Speed Limit

This setting defines a top speed for the Live Tooling. When this setting has a nonzero value, the Live Tooling will never exceed the designated speed.

355 - Sub Spindle Speed Limit

This setting defines a top speed for the Sub Spindle. When this setting has a nonzero value, the Sub Spindle will never exceed the designated speed.

356 - Beeper Volume

This setting allows the user to control the volume of the beeper in a control simulator. Setting a value of 0 will turn off the beeper.

357 - Warmup Compensation Cycle Start Idle Time

This setting defines an appropriate idle time, in hours, for warmup compensation to be restarted. When a machine has been idle longer than the amount of time in this setting, a **[CYCLE START]** will ask the user if he wishes to apply warmup compensation.

If the user answers with **[Y]** or **[ENTER]**, warmup compensation is applied a new, just as if the machine was powered up and **[CYCLE START]** commences. An **[N]** answer will continue cycle start with no warmup compensation. The next opportunity to apply warmup compensation will be after the setting 357 period has elapsed.